AZ-900 Final Exam Study Guide

Microsoft Azure Fundamentals — Quick Review Q&A, compiled from your practice session

1. Core Azure Architecture & Governance

Q: Which two components are created in an Azure subscription?

Answer: Resource groups and resources

Hierarchy: Management Groups → Subscriptions → Resource Groups → Resources. Management groups sit above subscriptions; Entra user accounts belong to the tenant, not the subscription.

Q: What is an Azure Storage account named storage001 an example of?

Answer: A resource

It lives inside a resource group. Subscription → Resource Group → Resource (storage001).

Q: Which Azure component allows you to replicate resources across a geography for business continuity during a natural disaster?

Answer: Region pairs

Scope size: Availability sets (rack/datacenter) → Availability zones (within one region) → Region pairs (across a geography, ~300+ miles apart).

Q: [Answer] is the deployment and management service for Azure.

Answer: Azure Resource Manager (ARM)

ARM is the layer every tool (Portal, CLI, PowerShell, SDKs, REST API) goes through to create/update/delete resources.

Q: Which two Azure resources can make use of availability zones?

Answer: Azure SQL databases and virtual machines

Zones apply to actual deployable workloads, not organizational containers like subscriptions or resource groups.

Q: Which Azure resource is a software emulation of a physical computer (virtual CPU, memory, storage, networking)?

Answer: A virtual machine

Abstraction gets lighter as you move: VM (full computer) → Container (shared OS kernel) → App Service/Functions (fully managed, no infra visibility).

Q: Which Azure compute service can you use to deploy and manage a set of identical virtual machines?

Answer: Azure Virtual Machine Scale Sets (VMSS)

Deploys/manages a load-balanced group of identical VMs, and can auto-scale the count up or down.

Q: What can you use to execute code in a serverless environment?

Answer: Azure Functions

You just write code; Azure handles all infrastructure, scaling, and billing per execution. (Logic Apps = serverless workflow/orchestration, not raw code execution.)

Q: What can you use to provide Mac and Android users access to a Windows environment to run Windows apps?

Answer: Azure Virtual Desktop

Streams a full Windows desktop/apps to virtually any device via client app or browser.

Q: Which management layer accepts requests from any Azure tool or API to create/update/delete resources?

Answer: Azure Resource Manager (ARM)

The consistent 'front door' behind every Azure interaction, regardless of which tool sends the request.

Q: Which two actions can be performed using the Azure portal?

Answer: Create new resources; Create a Microsoft Entra user

You cannot casually change an existing VM's availability zone (set at creation time) or assign specialized RBAC 'deny assignments' through normal portal UI.

2. Networking

Q: Which scenario is a use case for a VPN gateway?

Answer: Connecting an on-premises datacenter to an Azure virtual network

Encrypted site-to-site tunnel over the public internet, bridging on-premises and Azure.

Q: Which two services establish network connectivity between on-premises and Azure resources?

Answer: Azure VPN Gateway and ExpressRoute

VPN Gateway = encrypted tunnel over public internet. ExpressRoute = private, dedicated line via a connectivity provider (faster, more reliable, higher bandwidth, but costlier and slower to set up).

Q: S2S VPN: Which Azure object represents the on-premises network?

Answer: A local network gateway

Holds the on-prem VPN device's public IP + on-prem address ranges. The virtual network gateway represents the Azure side; a connection resource ties the two together.

Q: What can you use to restrict deployment of a VM to a specific location?

Answer: Azure Policy

Enforces rules/compliance (allowed locations, VM sizes, required tags). Different job than resource locks (which protect existing resources from deletion/changes).

3. Storage

Q: Which Blob storage tier stores data offline with the lowest storage cost and highest access cost?

Answer: Archive

Full order (most → least accessed): Hot → Cool → Cold → Archive. Storage cost and access cost move in opposite directions across all tiers.

Q: Which storage service offers fully managed cloud file shares accessible via SMB?

Answer: Azure Files

Also supports NFS (Linux, premium tier only). Azure Disk Storage is block storage for a single VM, not a shared network file system.

Q: Which storage service should you use for unstructured files (e.g., images) served on webpages?

Answer: Azure Blob storage

Object storage for massive unstructured data, accessible via HTTP/HTTPS URLs, often paired with a CDN.

Q: Which two protocols can be used to access Azure file shares?

Answer: SMB and NFS

SMB = cross-platform default. NFS = Linux-specific, premium file shares only. (HTTP/FTP are not supported for mounting shares.)

4. Identity, Access & Security

Q: What can you use to ensure a user can only access applications from compliant devices?

Answer: Conditional Access

If-then policy engine in Microsoft Entra. MFA is often an enforcement action WITHIN a Conditional Access policy, not a substitute for it.

Q: Which Microsoft Entra feature ensures users can only access Microsoft 365 apps from approved client applications?

Answer: Conditional Access

Conditions can include approved/compliant client apps, often paired with Intune app protection policies.

Q: What ensures users authenticate via MFA when signing in from a specific location?

Answer: Conditional Access

Classic use case: trigger MFA when sign-in occurs from an untrusted country/region or IP range.

Q: What can you apply to a VM to ensure users cannot delete the resource?

Answer: A lock (CanNotDelete)

Resource locks (CanNotDelete or ReadOnly) override even Owner/Contributor permissions. Separate entirely from RBAC.

5. Cloud Concepts, Cost Models & Deployment Models

Q: Which two attributes characterize the private cloud deployment model?

Answer: Hardware must be purchased; the company has complete control over physical resources and security

Quick provisioning/deprovisioning and pay-for-what-you-use are PUBLIC cloud traits, not private cloud.

Q: What are two characteristics of a consumption-based model?

Answer: No upfront costs; ability to stop paying for resources no longer used

High CapEx and owning/managing physical infrastructure describe the opposite model (CapEx/on-premises).

Q: Which two characteristics are common advantages of cloud computing?

Answer: Geo-distribution and high availability

Cloud enables (not eliminates) horizontal scaling, and removes (not grants) physical server access — that's the trade-off for outsourcing infrastructure.

Q: [Answer] refers to upfront, one-time costs such as hardware purchases.

Answer: Capital expenditures (CapEx)

CapEx = one-time, own the asset. OpEx = recurring, pay-as-you-go, rent the service. Cloud shifts spend from CapEx to OpEx.

Q: Increasing compute capacity by adding RAM/CPU to a VM is called...

Answer: Vertical scaling (scale up)

Vertical = bigger single machine. Horizontal = more machines (scale out).

Q: Deploying/configuring cloud resources quickly as business needs change is called...

Answer: Agility

Agility = speed of deployment. Elasticity = dynamic capacity matching demand. Scalability = capacity to grow. All distinct.

Q: Increasing compute capacity by adding instances (e.g., more VMs) is called...

Answer: Horizontal scaling (scale out)

Opposite approach from vertical scaling — distributes load across multiple machines instead of upsizing one.

Q: What are cloud-based backup services, data replication, and geo-distribution features of?

Answer: A disaster recovery plan

DR = 'what happens if things go wrong.' Elasticity/scalability = 'how we handle changing demand.' Two different problems.

Q: What is high availability in a public cloud environment dependent on?

Answer: The service-level agreement (SLA)

SLA is the formal, measurable uptime guarantee (e.g., 99.9%) that defines and underpins high availability.

Q: Which cloud service model is typically licensed through a monthly/annual subscription?

Answer: Software as a Service (SaaS)

SaaS = subscription-licensed, ready-to-use software. IaaS/PaaS are typically consumption/pay-as-you-go billed.

Q: In which cloud service model is the customer responsible for managing the operating system?

Answer: Infrastructure as a Service (IaaS)

Moving IaaS → PaaS → SaaS, the provider takes on more responsibility and the customer manages less.

Q: What is the customer responsible for in a SaaS model?

Answer: Data and access

Provider manages infrastructure, virtualization, OS, runtime, and the app itself. Customer just owns their data and who can access it.

Q: Which cloud service model is used by Microsoft 365?

Answer: Software as a Service (SaaS)

Anchor examples: Microsoft 365 = SaaS, Azure App Service = PaaS, Azure VMs = IaaS.

6. Management, Monitoring & Cost Tools

Q: You plan to build a PaaS solution. What should you use to estimate monthly costs?

Answer: Azure Pricing calculator

Forward-looking / pre-deployment estimate tool. Cost Management tracks actual spend after deployment; Advisor optimizes existing resources.

Q: You use [Target] to help organize Azure resources and track costs by department or project.

Answer: Tags

Flexible name-value metadata pairs. Azure Policy enforces rules/compliance, a different job entirely.

Q: You need a team of Linux administrators to manage Azure resources using Bash.

Answer: Azure CLI

Built around Bash-style syntax, cross-platform. Azure PowerShell uses cmdlet/object-pipeline syntax instead.

Q: What can you use to manage servers across third-party cloud platforms and on-premises environments?

Answer: Azure Arc

Extends Azure management (policies, governance, monitoring) to on-prem, AWS, GCP, and edge resources via one control plane.

Q: Two tools to create a new Azure VM from an Android mobile device?

Answer: The Azure portal and PowerShell in Azure Cloud Shell

Both run in-browser/no local install needed. RDP and SSH connect to VMs that already exist — they don't create new ones.

Q: Where should you look for a root cause analysis (RCA) / Post-Incident Review report for an outage?

Answer: Azure Service Health

Service Health = Azure's own service status dashboard: service issues (active problems), health advisories (retirements/breaking changes), planned maintenance, and PIR/RCA reports.

Q: You need to be notified about new recommendations for reducing Azure costs.

Answer: Azure Advisor

Personalized recommendations across cost, security, reliability, operational excellence, and performance.

Q: You need custom threshold-based autoscaling to scale an app up/down with demand.

Answer: Azure Monitor

Owns metrics collection + the autoscale engine. Application Insights (APM) is a feature within Azure Monitor, focused on app code telemetry.

Q: What should you proactively review to avoid service interruptions like retirements and breaking changes?

Answer: Health advisories

A category within Service Health: 'service issues' = active problems now; 'health advisories' = forward-looking changes to plan for.

Q: What can you use to automatically detect performance anomalies for web apps?

Answer: Azure Application Insights

Includes Smart Detection (ML-based) to flag anomalies like response-time spikes or unusual failure rates automatically.

Q: What can you use to get notified about an outage in a specific Azure region?

Answer: Azure Service Health

Personalized to the regions/services you actually use; supports outage alerts.

Q: Which service can generate an alert if VM utilization is over 80% for five minutes?

Answer: Azure Monitor

Metric-based alert rules with custom thresholds and time windows — one of Azure Monitor's most core, testable features.

Q: What can you use to find planned maintenance info for critical Azure services?

Answer: Azure Service Health

Same theme: Service Health owns everything about the status of Azure's OWN services (outages, advisories, maintenance).

Quick-Reference Contrast Tables

Scaling

Vertical Scaling	Horizontal Scaling
Bigger single machine (add RAM/CPU)	More machines (add VM instances)
Scale up/down	Scale out/in

Cost Models

CapEx	OpEx / Consumption-Based
One-time upfront cost	Recurring, pay-as-you-go
Own the asset (hardware)	Rent the service

Service Models (Responsibility)

IaaS	PaaS	SaaS
Customer manages OS, runtime, apps, data	Customer manages apps + data	Customer manages data + access only
Example: Azure VMs	Example: Azure App Service	Example: Microsoft 365

Connectivity

VPN Gateway	ExpressRoute
Encrypted tunnel over public internet	Private, dedicated connection via provider
Fast/cheap setup	Higher cost, higher bandwidth, more reliable

Monitoring & Health Tools

Tool	Purpose
Azure Monitor	Metrics, logs, alerts, autoscale for YOUR resources
Application Insights	APM: web app performance + anomaly detection (part of Monitor)
Azure Advisor	Personalized recommendations (cost/security/reliability/performance)
Azure Service Health	Status of AZURE'S OWN services: outages, advisories, maintenance

Availability Scope

Availability Sets	Availability Zones	Region Pairs
Hardware fault tolerance within one datacenter	Separate datacenters within one region	Separate regions within a geography

Good luck on the AZ-900 exam! Bon courage, rabi m3ak. This guide compiles the Q&A from your Claude practice session.